

MIDTERM EXAM

ECO403–INTRODUCTION TO CONTRACT THEORY & AUCTIONS

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Problem 1: More about hats

Aknur, Rysty, and Yerbol are standing first, second, and third in a line, respectively. Each one sees the persons standing in front of them. Prof. Alejandro announces: “I have three red hats and two white hats. I will place a hat on the head of each one of you.” After Prof. Alejandro places the hats, he asks Yerbol (who can see the hats worn by Rysty and Aknur) if he knows the color of the hat on his own head. Yerbol replies “no”. He then asks Rysty (who sees only Aknur’s hat) whether she knows the color of the hat on her own head, and she also replies “no”. Finally, he asks Aknur (who cannot see any of the hats) if she knows the color of the hat on her own head.

- 1.a Construct a set of states Ω that contains 7 states of the world and describes the situation above.
- 1.b What are the partitions of Aknur, Rysty, and Yerbol after Prof. Alejandro’s announcement and before he asked Yerbol whether he knows the color of his hat?
- 1.c What are the partitions of Aknur and Rysty after Yerbol’s response and before Rysty responded to Prof. Alejandro’s question?
- 1.d What is Aknur’s partition after hearing Rysty’s response?
- 1.e What is Aknur’s answer to Prof. Alejandro’s question? Does this answer depend on the state of the world, that is, on the colors of the hats that the three wear?

Problem 2: Who kidnapped Prof. Alejandro?

Professor Alejandro has disappeared without a trace! The students of ECON101 and ECON403 are two of the prime suspects in this academic whodunit. Why, you ask? Well, Professor Alejandro dared to give an ECON101 exam so tough that it left a trail of bruised egos and tears longer than an infinitely elastic demand curve! And as for ECON403, they had a midterm looming on the horizon. Witnesses also claim that Yernur, a past student from Prof. Alejandro, was caught yelling at him, “I’m furious! I was one of your best students, but my name never got to be the hero in one of your problem-solving adventures!”

Detective Nurdaulet and Detective Nurassyl are assigned to the case. They were trained at the same police academy. Their instruction consists of a well-defined rule specifying who to arrest given the clues that they have discovered. The two detectives are ordered to conduct an independent investigation. They promise not to share any data gathered from their research, and begin their sleuthing in different corners of NU. Suddenly the detectives are asked to appear and announce who they plan to arrest. Neither has had the time to complete a full investigation, so they each have gathered different clues. They meet on their way to the police station. Recalling their pledges, they do not tell each other a single discovery, or even a single reason why they were led to their respective conclusions. But they do tell each other who they plan to arrest. Hearing the other’s opinion, each detective may change his mind and give another opinion.

Suppose that the above situation and all of its specifics can be modeled with the following Belief space:

$$N = \{1, 2\}, \quad \Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$\mathcal{P}_1 = \{\{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\}\}, \quad \mathcal{P}_2 = \{\{1, 2, 3, 4\}, \{5, 6, 7, 8\}, \{9\}\}$$

$$\mu(\omega) = \frac{1}{9}, \quad \forall \omega \in \Omega.$$

Each $\omega \in \Omega$ represents all facets of the investigation related to the kidnapping. Detective Nurdaulet is player 1 and detective Nurassyl is player 2. Their information partitions represent the outcomes of their individual investigations. Consider the random variable $X : \Omega \rightarrow \mathbb{R}$:

ω	1	2	3	4	5	6	7	8	9
$X(\omega)$	17	-7	-7	0	17	-7	-7	-7	0

The different values of this random variable represent who is determined to be guilty at each state of the world: 17 for the ECON101 students, -7 for the ECON403 students and 0 for Yernur.

In addition to their training at the police academy, both Detective Nurdaulet and Detective Nurassyl are extremely smart and logical, they both know this, and they also took ECON403 with Prof. Alejandro. Suppose that Detective Nurdaulet and Detective Nurassyl talk long enough. Show that we can be sure that both detectives will announce the same suspect(s) at the station.

Problem 3: A Binary Investment Game

(Based on Bergemann and Morris 2016.) Consider the following investment game. Each player can either invest $a = I$ or not invest $a = N$, and the payoffs are given in the bad state θ_B and the good state θ_G by the matrices:

		Player 2	
		I	N
Player 1	I	$z - 1 + y_B, z - 1 + y_B$	$-1, z$
	N	$z, -1$	$0, 0$
		θ_B	
		Player 2	
		I	N
Player 1	I	$z + 1 + y_G, z + 1 + y_G$	$1, z$
	N	$z, 1$	$0, 0$
		θ_G	

The payoffs are symmetric and are composed of three components: (i) there is a payoff 1 to invest in the good state θ_G and a payoff of -1 to invest in the bad state θ_B ; (ii) there is always an externality $z > 0$ if the other player invests; (iii) there is an additional, possibly state dependent, payoff y_j , $j = B, G$, to investing if the other player invests as well. The payoff y_j can be positive or negative, but of uniform sign across states, leading to a game with strategic complements or substitutes, respectively.

- 3.a Suppose there is complete information, i.e., the players know in which state they are and that knowledge of the state is common knowledge. What conditions on z and y_j , $j = G, B$ need to be satisfied for invest to be a strictly dominant strategy in θ_G and not to invest to be a strictly dominant strategy in θ_B ?
- 3.b Assume that $z > 1$. What conditions on y_j , $j = G, B$ need to be satisfied for the sum of the payoffs for both players to be maximized when both players invest in both states?¹

From now on, assume that state θ_G occurs with probability ψ , while state θ_B occurs with probability $1 - \psi$, where $0 < \psi < 1$. Suppose that both players are uninformed about the state whether the state is good or bad. Assume also the conditions on the parameters in 3.(a) and 3.(b).

Suppose that there is an information technology π where, if the state is bad, each player observes a signal t_b with probability q . If a player does not receive the signal t_b , then he receives the signal t_g , and thus the signal t_g is always observed in the good state. To be more specific, below are the conditional distributions:

$\pi(t_1, t_2 \theta_B)$	t_b	t_g	$\pi(t_1, t_2 \theta_G)$	t_b	t_g
t_b	q	0	t_b	0	0
t_g	0	$1 - q$	t_g	0	1

¹Thus, in this question, I am asking for the conditions on the parameters that make investment from both players at all states welfare maximizing.

Thus q is a measure of the information technology's accuracy.

Let $T_i = \{t_b, t_g\}$, for $i = 1, 2$, and $\Theta = \{\theta_G, \theta_B\}$. Define the *set of states of the world* to be $\Omega = \Theta \times T_1 \times T_2$. Assume each player observes her own realization of the signal $t_i \in T_i$.

- 3.c Provide the belief space representation of the description above: list all the states of the world, specify the signalling functions τ_i , and compute the prior probability distribution μ over Ω , given ψ and $\pi(\cdot|\theta)$ as above. *Hint: use law of total probability, $\Pr(\theta, t_1, t_2) = \Pr(\theta) \times \Pr(t_1, t_2|\theta)$.*

Consider now the full Bayesian game with the belief space structure from item 3.(c).

- 3.d *Strategic Complements*. Assume that $y_j \geq 0$, $j = B, G$. What condition on the accuracy of the signal q needs to hold for “always invest” to be a Bayes-Nash Equilibrium?
- 3.e *Strategic Substitutes*. Assume that $y_j \leq 0$, $j = B, G$. What condition on the accuracy of the signal q needs to hold for “always invest” to be a Bayes-Nash Equilibrium?